

General Comments:

This manuscript explores the use of convolutional neural network models for estimating ages of chum salmon scales. The authors compiled an enormous dataset of chum scale images from populations across their range on the west coast of North America. They compared two image pre-processing techniques for three different CNN models, using a weighted average of the three CNNs as an “ensemble” to arrive at the final age estimates. Ensemble models under both pre-processing techniques demonstrated high agreement with human age readers, but the SAM method performed better overall.

I think this is an interesting paper on a new and relevant topic that is sure to be the direction salmon scale ageing is headed in the future. The manuscript is well written, and the conclusions supported by the results.

As a study about chum ages, this paper would benefit from some additional description in the methods of how chum age is defined. Additionally, a figure, or supplemental figure showing example images of the different age classes would be beneficial for the reader. Current scale image figures do not indicate age of the scales presented.

The main benefit outlined by authors of applying CNNs to scale aging is the potential for saving time and increased sample processing capacity for production ageing labs. The major limitation to this would be the actual digitization of the scales into images. With the equipment we have in our lab, we can manually age chum scales significantly faster than we can take images of those same scales. Authors briefly mention this in the discussion, but it would be nice to see this expanded and potential solutions (e.g. automated imaging software) mentioned.

Specific Comments:

Abstract

Line 27 – Please indicate the different CNNs that were used in the ensemble (at the least, how many) and that you used a weighted average to create the ensemble.

Line 30 – Report agreement for both pre-processing techniques, not just the better one.

Introduction

Line 109 – improve, not increase

Line 117 – please indicate what you mean by “origin.” Hatchery vs. wild?

Methods

Line 130 – Somewhere in this “Scale Datasets” section indicate how ages were defined. I.e. is this the number of annuli present on the scale, or the total age including the winter during egg incubation (Run/capture year – brood year)? Are you using a January 1st birthday? Because some samples are from immature fish captured in the ocean, date of capture is relevant.

Line 168 – No indication of the magnification at which these various collections were imaged at. Please provide if available. Indicate if magnification is consistent or variable. Does magnification become irrelevant because of the SAM/YOLO preprocessing? If so, state that.

Line 176 – What was your subsampling scheme? Please describe how you decided how many images from each population’s more common age classes to exclude/include?

Results

Line 352 – Please describe directional bias present in the model age estimates. Yes, most disagreements were +/- 1 year, but younger fish tended to be over-aged and older fish tended to be underaged by the models.

Line 379 – 70% vs. 87% agreement for Chum vs. Atlantic salmon seems like a huge difference, particularly in the context of salmon ageing where these ages may go on to be used in survival and forecasting models where that error will compound. Why do you describe this difference as “slightly lower agreement” ?

Line 383 – Again, 70% vs. 91% and 89% seems like huge differences in agreement. Why do you describe the performance of the older -B4 model as only “slightly worse” ?

Discussion

Line 407 – Why do you think model performed well on age-1 fish despite having very few age-1 samples?

Line 452 – add the word “may” to : “because they are similar in appearance and may result in the overestimation of age...”

Line 517 – I think this is a key point worth expanding on. With the equipment set up in our lab (and other age labs I have visited), digitizing chum scales takes significantly longer than just manually reading and producing age estimates. This bottleneck would make the image processing efficiency of the CNN models irrelevant in the setting of a 24 hour turn around window. However, companies like Zeiss now have automated imaging software that can map and image all scales on a slide/acetate which would significantly speed up digitization process.

Line 530 – To my knowledge, H/W calls based on scale patterns are only reliable for fish that spend ≥ 1 year in freshwater, which Chum do not. Even then the ability to differentiate H/W, and the criteria used is typically population specific. Have you tried this using the CNNs with Chum? This would be very cool if it works but I would be surprised.

Line 533 – Scale patterns are just a reflection of growth. Yes, there are differences in growth conditions by region/basin, but interannual differences within regions/basins can also be significant. Chum salmon leave their natal freshwater habitat and enter the marine environment days/weeks after hatching. This migration is happening around the same time scales are just beginning to form. Identifying basin of origin based on scale pattern is a very lofty goal, especially considering at maximum only a handful of circuli could have formed by the time a chum leaves the freshwater environment. Saying that this is information that could be extractable by the models is too strong.

Figure and Tables

Line 745 – Please state that the table indicates the number of discrepancies.